TIG Energy Arbitrage
Parameter Calibration Report

Experimental Results

March 4, 2026

1 Overview

This report documents the parameter calibration for the TIG Energy Arbitrage Level 2 challenge across five tracks of increasing difficulty. The goal is to find parameter ranges that produce:

- Positive, increasing decomposition profit across tracks
- Meaningful greedy-to-decomposition gap that increases with track difficulty
- 100% feasibility for both solvers
- Profit CV in $[0.1, 0.7]$
- Fast verification (< 50 ms)
- Meaningful congestion frequency

1.1 Modified Parameters

Starting from the original specification defaults, the following parameters were tuned:

Table 1: Parameter changes from original specification

Parameter	**Original**	**Tuned**	**Rationale**
F_{base} (`NOMINAL_FLOW_LIMIT`)	100 MW	50 MW	Tighter limits for battery impact
τ_{cong} (`TAU_CONG`)	0.97	0.90	Earlier congestion detection
p_{base} (`base_load`)	50 MW	200 MW	Higher exo injections
γ_{pat} (`pattern_scale`)	20	60	Stronger temporal patterns
γ_{noise} (`noise_scale`)	2	5	More injection variability
ϕ_{margin} (`flow_margin`)	0.70	0.85	Tighter exo flow packing
Rescaling	Global	Per-timestep	Avoids one-bad-step dilution

The key insight was that the original global rescaling approach found the single worst (line, timestep) pair and scaled *all* injections down by that factor, resulting in only 5–9% average line utilization. Per-timestep rescaling ensures the binding line per timestep reaches $\phi_{\text{margin}} = 85\%$, while the average across all lines is 17–28% (due to PTDF flow distribution).

2 Track Parameters

Table **??** shows the five track parameter regimes from the specification.

Table 2: Track parameter regimes

Track	n	L	m	H	γ_{cong}	σ	ρ_{jump}	α	h
1	20	30	10	96	1.00	0.10	0.01	4.0	0.2
2	40	60	20	96	0.80	0.15	0.02	3.5	0.4
3	80	120	40	192	0.60	0.20	0.03	3.0	0.6
4	100	200	60	192	0.50	0.25	0.04	2.7	0.8
5	150	300	100	192	0.40	0.30	0.05	2.5	1.0

3 Main Results: Track Progression

All experiments use 10 random seeds per track. Table **??** summarizes the key metrics.

Table 3: Track progression results (10 seeds each)

Track	**Greedy $**	**Decomp $**	**Gap**	**Cong%**	**Active%**	**CV**	**De ms**	**V ms**
1	16 626	19 354	14%	31.4	46.5	0.35	27	1.2
2	30 027	38 708	22%	32.1	44.9	0.19	210	2.4
3	44 319	80 145	45%	30.5	26.5	0.31	3 794	11.5
4	67 411	138 181	51%	42.4	24.6	0.17	12 168	18.5
5	49 010	182 237	73%	68.8	40.6	0.31	16 550	21.4

3.1 Key Observations

- **Gap increases with track difficulty**: 14% → 22% → 45% → 51% → 73%. This confirms that larger, more constrained networks require increasingly sophisticated algorithms.
- **Decomposition profit increases**: \$19K → \$39K → \$80K → \$138K → \$182K, reflecting more batteries and longer horizons.
- **100% feasibility** for both solvers across all tracks.
- **CV in [0.17, 0.35]**: healthy variability, neither too deterministic nor too noisy.
- **Verification time**: 1–22 ms, well within the TIG fast-verification requirement.
- **Solve time scales appropriately**: 27 ms (Track 1) to 16.5 s (Track 5), creating computational difficulty.
- **Congestion**: 30–69% of steps, driven by network constraint tightness (γ_{cong}).

4 Visualizations

4.1 Greedy–Decomposition Gap Across Tracks

See Figure **??**

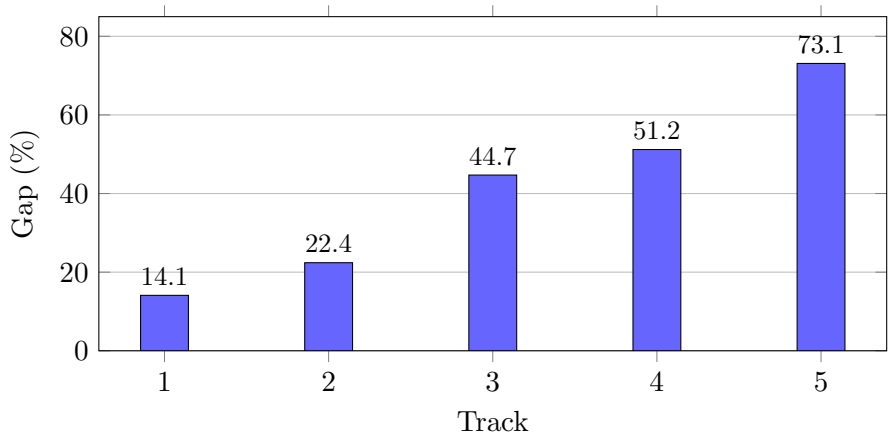

Figure 1: Greedy-to-decomposition profit gap by track. The gap increases from 14% (Track 1) to 73% (Track 5), showing that network complexity drives algorithmic difficulty.

4.2 Profit Comparison

See Figure **??**

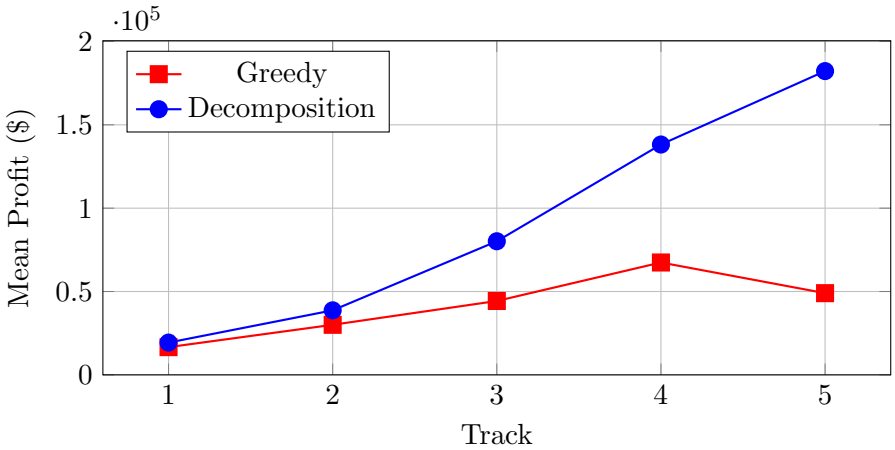

Figure 2: Mean profit by track. Decomposition profit scales with problem size. Greedy profit plateaus for Track 5 due to tight flow constraints ($\gamma_{\text{cong}} = 0.40$, effective limit 20 MW).

4.3 Congestion and Flow Utilization

See Figure **??**

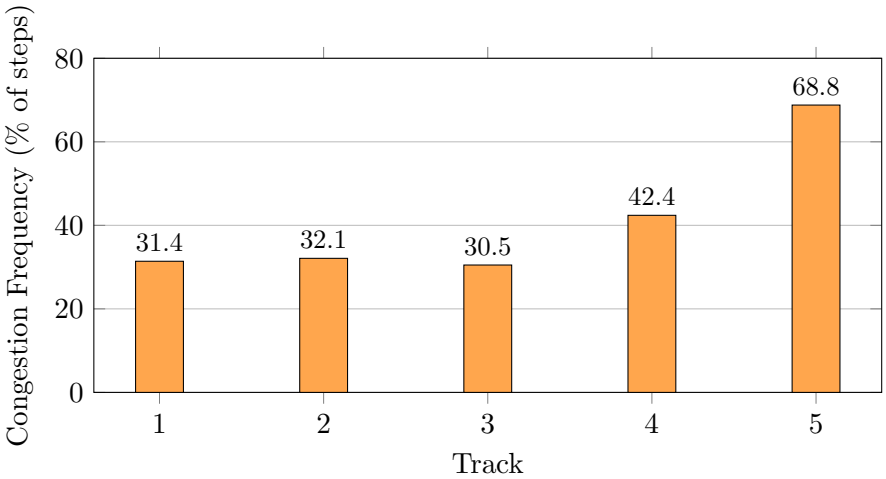

Figure 3: Congestion frequency (fraction of timesteps with $\max_l |f_l|/\bar{F}_l \geq \tau_{\text{cong}}$). Congestion is generally higher for tracks with tighter γ_{cong} (Tracks 4–5), though solver strategy also influences this metric.

4.4 Solve and Verification Time

See Figure **??**.

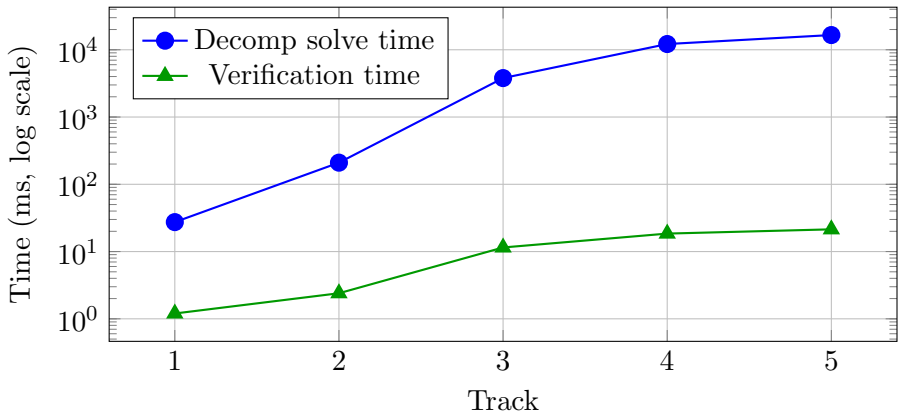

Figure 4: Solve vs. verification time. The "hard-to-solve, easy-to-verify" property holds: solve time grows 600× from Track 1 to Track 5, while verification stays under 25 ms.

4.5 Profit Distribution

See Figure **??**

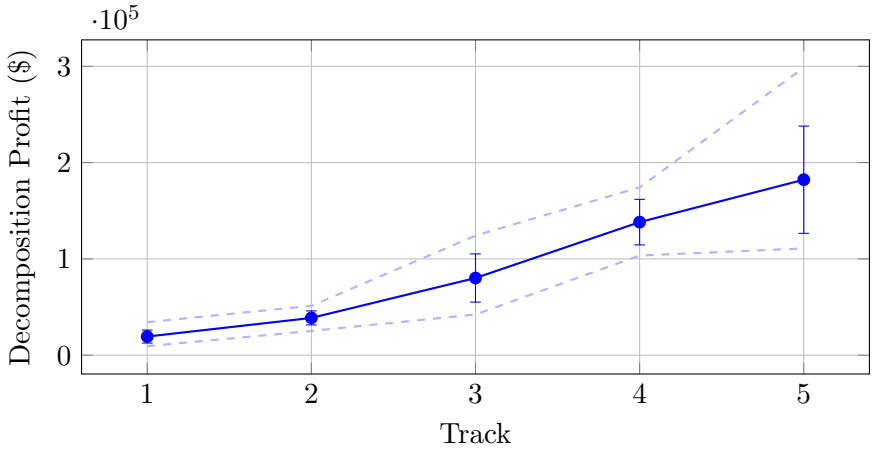

Figure 5: Decomposition profit distribution: mean ± 1 std (error bars). Dashed lines show min and max across 10 seeds. The spread increases with track size, reflecting more stochastic instances.

5 Congestion Profile Analysis

Table **??** shows the fraction of timesteps where the maximum line utilization exceeds various thresholds, measured under the decomposition solution.

Table 4: Congestion profile: % of steps with $\max_l |f_l|/\bar{F}_l \geq \tau$ (under decomp solution)

Track	**Eff. Limit**	$\tau = 0.70$	$\tau = 0.80$	$\tau = 0.85$	$\tau = 0.90$	$\tau = 0.95$	$\tau = 0.97$
1	50 MW	95.8%	91.2%	65.4%	32.7%	17.5%	3.1%
2	40 MW	90.6%	85.6%	71.0%	40.0%	17.3%	9.6%
3	30 MW	97.9%	95.7%	70.8%	29.0%	20.9%	17.9%
4	25 MW	97.9%	94.8%	70.9%	34.0%	24.7%	21.0%
5	20 MW	96.0%	92.4%	86.1%	71.5%	54.8%	48.0%

At $\tau_{\text{cong}} = 0.90$, congestion ranges from 29% (Track 3) to 72% (Track 5). At the original threshold $\tau = 0.97$, congestion ranges from 3% to 48%, with Track 5 showing substantial near-limit operation.

5.1 Per-Line Utilization Distribution

Under idle (exogenous-only) dispatch, the per-timestep scaling ensures the binding line reaches exactly $\phi_{\text{margin}} = 85\%$. However, the fraction of *all* lines above various thresholds decreases with network size due to PTDF flow distribution:

Table 5: Fraction of (line, timestep) pairs above utilization thresholds (exo-only)

Track	**Lines**	$> 50\%$	$> 70\%$	$> 80\%$	$> 90\%$
1	30	15.7%	6.4%	4.2%	0.0%
2	60	11.1%	4.0%	2.3%	0.0%
3	120	7.7%	2.3%	1.2%	0.0%
4	200	5.1%	1.4%	0.7%	0.0%
5	300	3.4%	0.9%	0.5%	0.0%

This shows that while the binding line per timestep is at 85%, larger networks distribute flows more broadly via the PTDF matrix, so fewer lines are near their limits on average.

6 Parameter Sensitivity

6.1 Congestion Factor γ_{cong}

Sweeping γ_{cong} within Track 1 (5 seeds, other parameters fixed, cf Figure **??**):

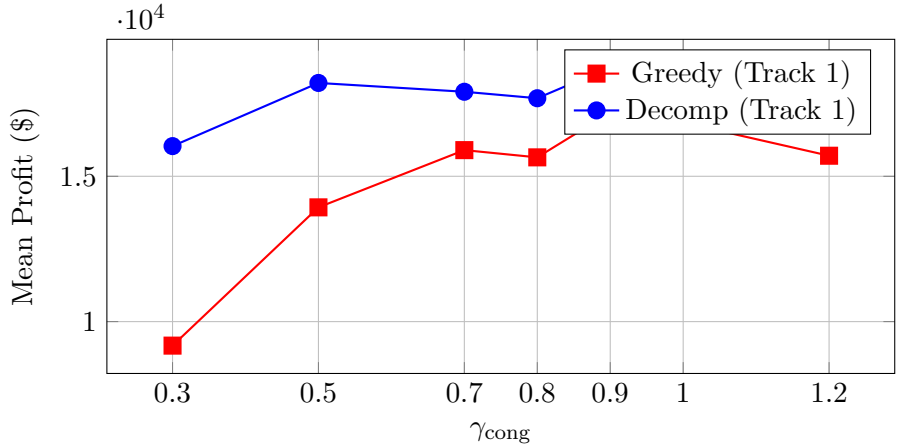

Figure 6: Effect of γ_{cong} on profit (Track 1). Very tight limits ($\gamma_{\text{cong}} = 0.3$) reduce greedy profit substantially while decomp remains robust. Optimal range: $\gamma_{\text{cong}} \in [0.5, 1.0]$.

For Track 2 (see Figure **??**):

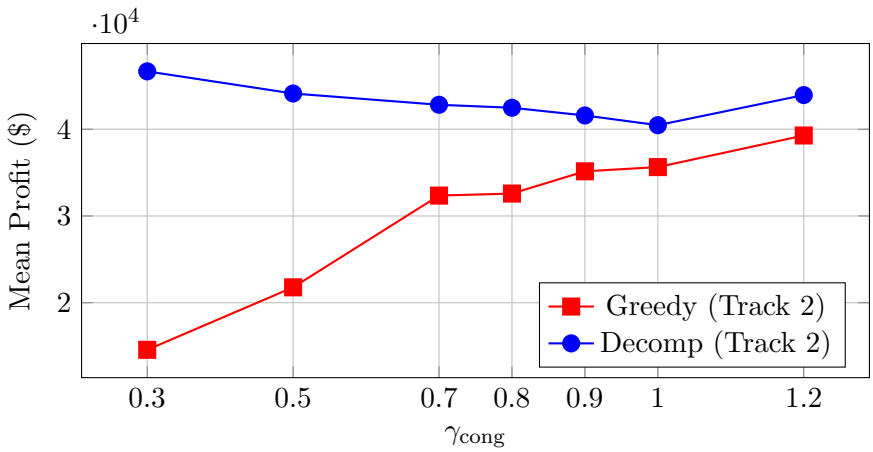

Figure 7: Effect of γ_{cong} on profit (Track 2). Greedy profit increases monotonically with looser limits, while decomp profit is relatively stable — the gap narrows from 69% ($\gamma_{\text{cong}} = 0.3$) to 10% ($\gamma_{\text{cong}} = 1.2$).

6.2 Volatility σ

See Figure **??**

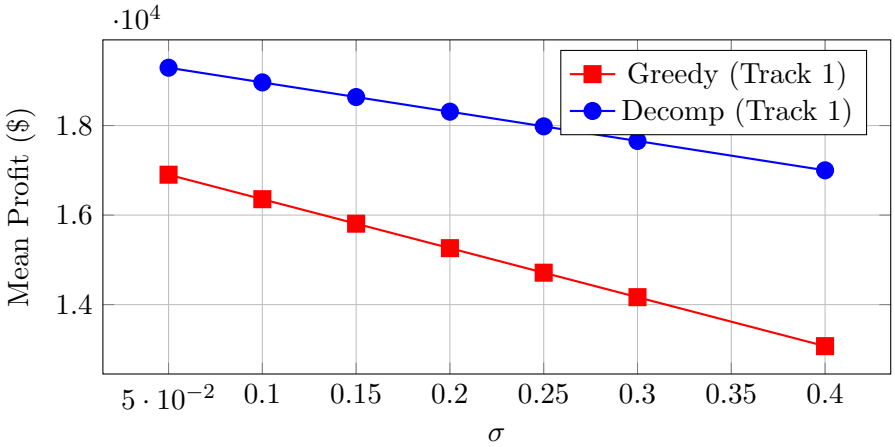

Figure 8: Effect of volatility σ on profit (Track 1). Both solvers show mild, near-linear profit decline with increasing volatility. The gap is stable at ∼14%.

6.3 Battery Heterogeneity h

See Figure **??**

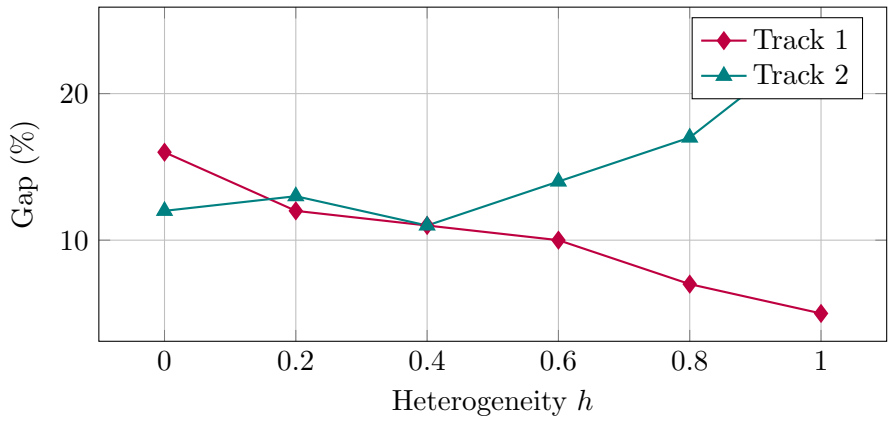

Figure 9: Effect of battery heterogeneity h on gap. For Track 1 (small network), higher heterogeneity *decreases* the gap (greedy benefits from diverse batteries). For Track 2 (larger network), higher heterogeneity *increases* the gap (coordination across diverse batteries becomes harder).

7 Summary of Parameter Ranges

Table **??** summarizes the observed metric ranges across all five tracks.

8 Conclusions

The calibrated parameter set produces a well-structured difficulty progression:

1. **Track 1** ($n = 20$, $\gamma_{\text{cong}} = 1.0$): Baseline correctness track. Small network with loose constraints. Gap is low (14%) — even a naive greedy captures most value through temporal arbitrage. Solve time: 27 ms.

2. **Track 2** ($n = 40$, $\gamma_{\text{cong}} = 0.8$): Moderate congestion emerges. Gap increases to 22% as tighter limits start constraining the greedy. Solve time: 210 ms.

Table 6: Metric ranges across tracks

Metric	**Range**	**Target**
Greedy–Decomp Gap	14–73%	$> 0\%$, increasing
Decomp Profit	\$19K–\$182K	Positive, increasing
Greedy Profit	\$17K–\$67K	Positive
Feasibility	100%	100%
Profit CV	0.17–0.35	$[0.1, 0.7]$
Verification Time	1–22 ms	< 50 ms
Decomp Solve Time	27 ms–16.5 s	Increasing
Congestion Frequency	30–69%	Meaningful
Effective Line Limits	20–50 MW	—

3. **Track 3** ($n = 80$, $\gamma_{\text{cong}} = 0.6$): The "transition" track where network effects dominate. Gap jumps to 45% — the greedy can no longer capture half the available value. Solve time: 3.8 s.

4. **Track 4** ($n = 100$, $\gamma_{\text{cong}} = 0.5$): Dense network with heavy tails ($\alpha = 2.7$). Gap at 51% with 42% congestion frequency. Risk management becomes critical. Solve time: 12 s.

5. **Track 5** ($n = 150$, $\gamma_{\text{cong}} = 0.4$): Full-scale challenge with maximum heterogeneity ($h = 1.0$). Gap reaches 73% — only sophisticated network-aware optimization can extract value. Congestion at 69% of steps. Solve time: 16.5 s.

The "hard-to-solve, easy-to-verify" property is well satisfied: solve time grows 600× from Track 1 to Track 5 while verification remains under 25 ms throughout.